Task 5: Dynamic Quoting Under Inventory Pressure

Approach and functional form

The submitted strategy is a dynamic inventory-aware quoting rule from the LP perspective. The required function **quote(inventory, sigma, alpha, eta)** returns relative bid and ask half-spreads. Here, **sigma** is realized volatility, **alpha** is the Task 3 adversity score, and **eta** is the fraction of the trading day elapsed.

The quote starts with a volatility-based spread and widens only when predicted adversity is above 0.5:

```
base = C_base * sigma     risk_multiplier = 1 + K_alpha * max(alpha - 0.5, 0)

half_spread = base * risk_multiplier
```

Then an inventory skew is added. If inventory is positive, the LP is long, so the bid is widened and the ask is tightened. If inventory is negative, the LP is short, so the bid is tightened and the ask is widened.

To avoid overreacting to small positions, the final rule uses an inventory deadzone:

```
adjusted_inventory = sign(inventory) * (abs(inventory) - inventory_deadzone)

inventory_pressure = tanh(adjusted_inventory / inventory_scale)

time_pressure = 1 + K_eta * eta^2

skew = K_skew * sigma * inventory_pressure * time_pressure

delta_bid = half_spread + skew     delta_ask = half_spread - skew
```

Both outputs are clipped to the contest bounds: **0.5 * sigma <= delta_bid, delta_ask <= 0.005**.

Intuition behind the chosen parameters

The final parameters were **C_base=1.0, K_alpha=1.0, inventory_scale=400, K_skew=0.60, K_eta=2.0, inventory_deadzone=75**. The idea is to keep quotes normal in calm conditions, widen for clearly adverse flow, and only push inventory down when the position becomes meaningful. The time pressure term makes the skew stronger near close, because ending the day with high inventory is risky.

An earlier version skewed continuously. It controlled inventory but was too defensive and gave up PnL in calm periods. The final version ignores small inventory and reacts more strongly only when inventory pressure or closing-time risk is important.

Validation methodology and results

Since no separate Task 5 training dataset was provided, the supplied trade data was used as a proxy backtest environment. Data was sorted chronologically, and the latest 20% of dates were kept as the held-out test split. Realized volatility used only previous mid-price returns within the same day, so future information was not used.

Strategy	Net PnL	Sharpe-like	Max DD	Avg End Inv
Fixed	9419.47	24.83	20.87	1454.89
Vol + Alpha	9510.79	26.07	10.33	1144.63
Inventory Skew	9516.91	26.77	2.28	119.70

The final inventory-skew strategy gave the best overall result. It slightly improved net PnL and Sharpe-like score, but the main improvement was risk control: average end-of-day inventory fell from more than 1100 units to about 120 units, and max drawdown reduced sharply. This makes the rule more stable because it earns PnL while also controlling inventory risk near close.